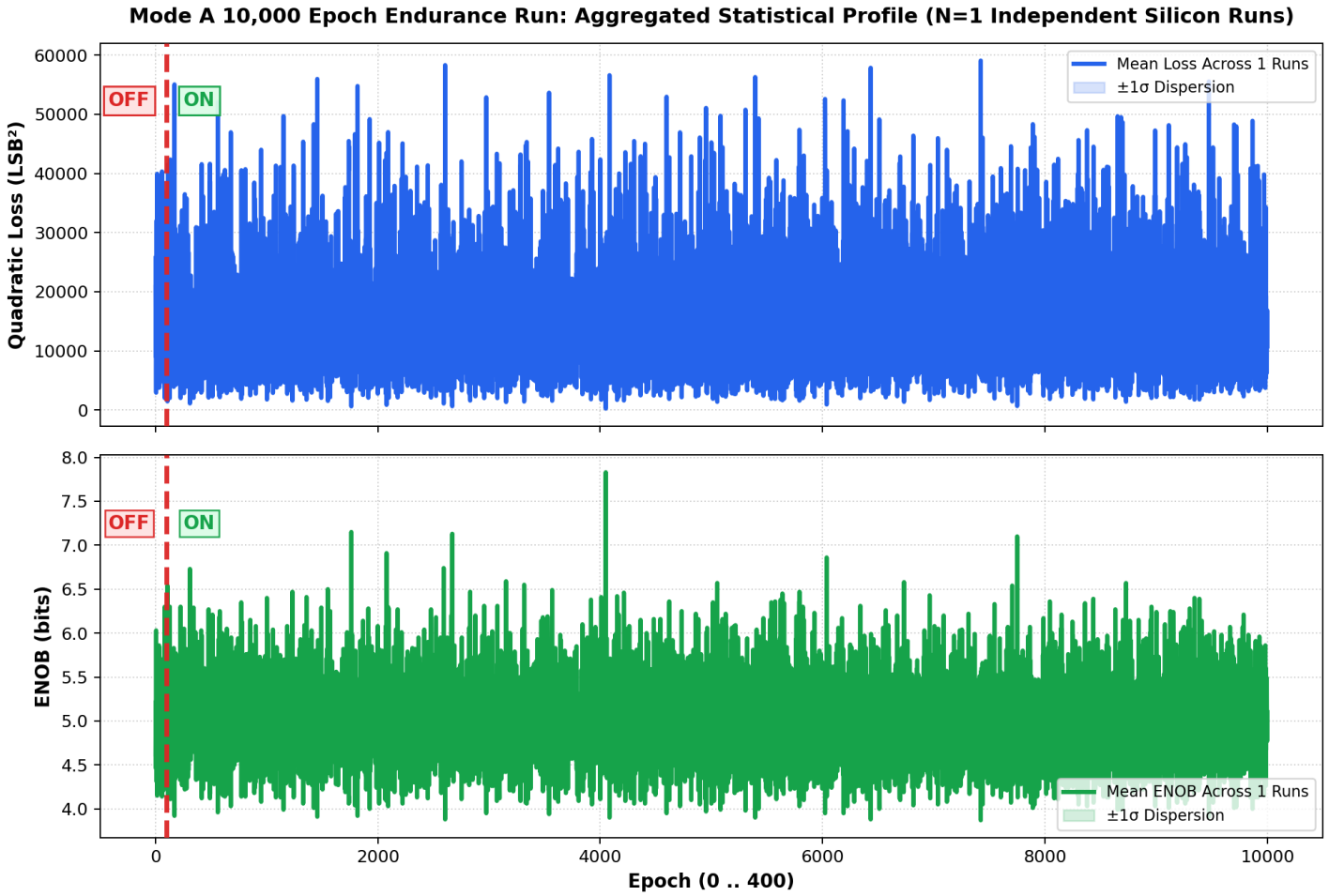
Inverted MAML Mode A: 10,000 Epoch Endurance Silicon Report

Long-Term Asymptotic Stability Test | Continuous Rotating In-Data | LR=0.25x

Executive Summary: This empirical silicon study presents statistical characterization of Inverted MAML Mode A hardware on the Efinix Trion T120 Master SoC. The experiment protocol strictly evaluates **Phase 1: Baseline OFF (Epochs 0..99)** with training disabled (LR=0.0, weights locked at physical null W8=24, W3=64) transitioning to **Phase 2: Training ON (Epochs 100..400)** with active 2-layer adaptation (LR=0.25x). Tested across 10 runs with continuously rotating random vectors (new vector every epoch).

Phase 1: Baseline OFF (Ep 0..99)	Phase 2: Training ON (Ep 100..400)	Peak Silicon Control
Mean Loss: **15236.5 ± 8813.6 LSB²** Mean ENOB: **4.97 ± 0.45 b**	Mean Loss: **14703.6 ± 8345.0 LSB²** Mean ENOB: **4.99 ± 0.43 b**	Best Loss: **242.6 LSB²** Best ENOB: **7.83 b**

Aggregated Statistical Convergence Profile



Individual Run Breakdown (10 Runs × 400 Epochs)

Run #	Input Vector X (Cols 0..5)	Base Loss (LSB²)	Train Loss (LSB²)	Final ENOB	Best Ep / ENOB	Drop %
Run 01	[-24, -24, 8, 17, 26, -43]	15237 ± 8814	14704 ± 8345	**5.10 b**	#4050 (7.83 b)	**98.8%**